Practice on Pattern Questions (Loops)

Practice Questions

1. Right Triangle Pattern

Problem: Print a right triangle pattern using asterisks (*).

```java
public class RightTriangle {
    public static void main(String[] args) {
        int n = 5;
        for (int i = 1; i <= n; i++) {
            for (int j = 1; j <= i; j++) {
                System.out.print("* ");
            }
            System.out.println();
        }
    }
}
```

Output:

```
*
* *
* * *
* * * *
* * * * *
```

2. Inverted Right Triangle Pattern

Problem: Print an inverted right triangle pattern using asterisks (*).

```java
public class InvertedTriangle {
    public static void main(String[] args) {
        int n = 5;
        for (int i = n; i >= 1; i--) {
            for (int j = 1; j <= i; j++) {
                System.out.print("* ");
            }
            System.out.println();
        }
    }
}
```

Output:

```
* * * * *
* * * *
* * *
* *
*
```

3. Number Pattern (Incrementing)

Problem: Print a number pattern where each row contains incrementing numbers.

```java
public class NumberPattern {
    public static void main(String[] args) {
        int n = 4;
        for (int i = 1; i <= n; i++) {
            for (int j = 1; j <= i; j++) {
                System.out.print(j + " ");
            }
            System.out.println();
        }
    }
}
```

Output:

```
1
1 2
1 2 3
1 2 3 4
```

4. Square Pattern

Problem: Print a square pattern using asterisks (*).

```java
public class SquarePattern {
    public static void main(String[] args) {
        int n = 4;
        for (int i = 1; i <= n; i++) {
            for (int j = 1; j <= n; j++) {
                System.out.print("* ");
            }
            System.out.println();
        }
    }
}
```

Output:

```
* * * *
* * * *
* * * *
* * * *
```

5. Hollow Square Pattern

Problem: Print a hollow square pattern using asterisks (*).

```java
public class HollowSquare {
    public static void main(String[] args) {
        int n = 5;
        for (int i = 1; i <= n; i++) {
            for (int j = 1; j <= n; j++) {
                if (i == 1 || i == n || j == 1 || j == n) {
                    System.out.print("* ");
                } else {
                    System.out.print("  ");
                }
            }
            System.out.println();
        }
    }
}
```

Output:

```
* * * * *
*       *
*       *
*       *
* * * * *
```

6. Floyd's Triangle

Problem: Print Floyd's triangle (consecutive numbers in triangular form).

```java
public class FloydTriangle {
    public static void main(String[] args) {
        int n = 4;
        int num = 1;
        for (int i = 1; i <= n; i++) {
            for (int j = 1; j <= i; j++) {
                System.out.print(num + " ");
                num++;
            }
            System.out.println();
        }
    }
}
```

Output:

```
1
2 3
4 5 6
7 8 9 10
```

7. Pyramid Pattern

Problem: Print a pyramid pattern using asterisks (*).

```java
public class PyramidPattern {
    public static void main(String[] args) {
        int n = 4;
        for (int i = 1; i <= n; i++) {
            // Print spaces
            for (int j = 1; j <= n - i; j++) {
                System.out.print(" ");
            }
            // Print stars
            for (int k = 1; k <= i; k++) {
                System.out.print("* ");
            }
            System.out.println();
        }
    }
}
```

Output:

```
   *
  * *
 * * *
* * * *
```

8. Diamond Pattern

Problem: Print a diamond pattern using asterisks (*).

```java
public class DiamondPattern {
    public static void main(String[] args) {
        int n = 4;
        // Upper half
        for (int i = 1; i <= n; i++) {
            for (int j = 1; j <= n - i; j++) {
                System.out.print(" ");
            }
            for (int k = 1; k <= i; k++) {
                System.out.print("* ");
            }
            System.out.println();
        }
        // Lower half
        for (int i = n - 1; i >= 1; i--) {
            for (int j = 1; j <= n - i; j++) {
                System.out.print(" ");
            }
            for (int k = 1; k <= i; k++) {
                System.out.print("* ");
            }
            System.out.println();
        }
    }
}
```

Output:

```
   *
  * *
 * * *
* * * *
 * * *
  * *
   *
```

Quiz Questions

1. Which loop is most commonly used for pattern printing in Java? a) while loop b) do-while loop c) for loop d) enhanced for loop

Answer: c) for loop

2. In a right triangle pattern with 5 rows, how many stars will be printed in the 3rd row? a) 2 b) 3 c) 4 d) 5

Answer: b) 3

3. What does the outer loop typically control in pattern printing? a) Number of columns b) Number of rows c) Number of spaces d) Number of characters

Answer: b) Number of rows

4. To create a hollow square pattern, which positions should contain stars? a) All positions b) Only corners c) Border positions only d) Center positions only

Answer: c) Border positions only

5. In Floyd's triangle, if you start with number 1, what number appears at the beginning of the 4th row? a) 6 b) 7 c) 8 d) 10

Answer: b) 7

Practice Coding Questions

1. Inverted Pyramid Pattern Write a Java program to print an inverted pyramid pattern:

```
* * * *
 * * *
  * *
   *
```

2. Number Triangle (Row Number Pattern) Write a Java program to print the following pattern:

```
1
2 2
3 3 3
4 4 4 4
```

3. Butterfly Pattern Write a Java program to create a butterfly pattern using stars (*) with 4 rows in the upper half.

4. Hollow Right Triangle Write a Java program to print a hollow right triangle pattern where only the borders contain stars and the inside is empty.

5. Pascal's Triangle (Numbers) Write a Java program to print Pascal's triangle with numbers for n=5 rows.